

Impacts of *Dobbs v. Jackson* on Medical Residency Training

Introduction

On June 24th, 2022, the U.S. Supreme Court abolished the federal right to abortion in *Dobbs v. Jackson Women's health*, endowing states with nearly unlimited regulatory authority over abortion for the first time in almost fifty years. No longer inhibited by precedent established in previous cases including *Roe v. Wade* and *Planned Parenthood v. Casey*, states adopted laws ranging from statutory and constitutional protections to gestational age limits and complete bans on abortion.¹ The resulting patchwork system of abortion access could affect individual preferences for living in a state where abortion is illegal, as people weigh the implications for personal autonomy, privacy, and other factors. Abortion restrictions carry additional considerations for those practicing and training in reproductive healthcare, who face an additional layer of legal scrutiny in states where abortion is illegal, and whose practice location decisions ultimately determine the availability of reproductive and maternal health services.

The labor market for new physicians (also known as the National Residency Match) is a natural setting to empirically assess these implications for two key reasons: (1) residency applicants are a highly mobile population,^{2,3} and (2) the market opens and closes on a discrete timeline each year. These factors make any effects from the adoption of an abortion restriction much clearer than in other settings, where individual circumstance typically determines when and if it is possible to enter the labor market, and factors like one's current employment or the needs of family might delay response. Additionally, residents are a key source of care for vulnerable populations, often having

high shares of Medicaid and charity care cases in hospital settings and as staff for community and student-run clinics, and the location of residency training is highly predictive of future practice patterns, so impacts on resident labor could shape provider availability in both the short and long-term. In this study, we assess whether state abortion restrictions enabled by *Dobbs* had an effect on the supply of new physicians to residency training programs.

How Abortion Restrictions Influence Labor Markets

Abortion protections established in *Roe v. Wade* over fifty years ago were associated with greater earnings, labor force participation, occupational prestige, and educational attainment among women.^{4,5,6,7,8,9,10,11} These factors have been hypothesized to be driven largely by the ability to exercise greater control over fertility decisions afforded by abortion protections. For instance, delaying childbirth until after completing a degree is made more difficult under restrictive reproductive health policies, and exercising control over the size of one's family (i.e., number of children) can be challenging in an environment in which reproductive rights are limited. Though these factors can also be determined by access to contraceptives, prior research has demonstrated that abortion access plays a pivotal role mediating reproductive autonomy, even when contraceptives are available.¹²

How Abortion Restrictions Affect Providers

States with abortion restrictions enabled by *Dobbs* have seen increases in infant and maternal mortality and morbidity, and reports of lower quality and reduced access to

obstetric and gynecological care unrelated to abortion following the adoption of the policy.^{13,14,15} Ob-gyns practicing in these states have reported that broadly written policy creates uncertainty over which services are permissible under abortion bans, with providers either delaying treatment until the legality of treatment can be determined or providing non-standard care to sidestep care options that are more likely to be scrutinized.^{16,17} In doing so, providers expose themselves to greater malpractice risk, creating a double bind between civil and criminal legal risk.¹⁸

These factors can also create spillover consequences for patients not specifically seeking an abortion, with research showing restrictive abortion policy has worsened the management of miscarriage, ectopic pregnancy, and early-pregnancy hemorrhage, increased the frequency of obstetric-related emergencies, and has led to broad reductions in the quality of emergency care.^{19,20,21,22,23} Some restrictions can also directly impact patients' friends, families, and healthcare providers, with several states adopting abortion "bounty hunter" laws that endow private citizens with the authority to sue anyone who assists with an abortion.²⁴ For reproductive healthcare providers, not only can restrictive abortion policy affect their practice, but it can also impact access to safe and high quality maternal health services for both themselves and their loved ones, which may influence where they decide to work, train, and reside.

Why Providers May Prefer to Work and Train in a Restrictive State

While there are several costs associated with residence in a state that has restricted abortion, preference for where one lives and one's beliefs surrounding abortion are

complex, varied, and individual. Around a third of individuals believe abortion should be illegal, and job applicants in this group may prefer to live in a state where that is the case.²⁵ Roughly 7% of respondents to a survey assessing physician and trainee practice location reported a preference for abortion-restrictive states, and roughly 8% said *Dobbs* had no bearing on their practice location decision. Preferences for practicing in a restrictive state may stem from personal beliefs but could also be motivated by the desire to reduce unmet need, given evidence of diminishing access to quality maternal health services in restrictive states.^{26,27,28} Potentially demonstrative of this possibility, national abortion counts have increased since *Dobbs*, in part because providers are serving an increasing share of patients located in restrictive states, i.e. through telehealth and out-of-state travel, despite direct legal threat from states attempting to enforce their restrictions.^{29,30,31,32}

Evidence of Dobbs' Influence on the Provider Workforce

A survey of provider practice location preferences following *Dobbs* indicates that most current physicians and trainees would prefer to work in a state where abortion is legal, citing concerns over their patients' access to comprehensive treatment, access for themselves and their families, and personal beliefs around bodily autonomy, regardless of their current state of residence.³³ Another survey found that roughly one in five ob-gyn residents had changed their future practice location plans because of *Dobbs*.³⁴ Evidence on the realized supply of physicians to restrictive states is mixed. There is evidence that fewer ob-gyn residents matched to training programs following *Dobbs*,³⁵ however other research indicates there was no change in resident supply, as a

reduction in residents graduating from medical schools located in states that protect abortion was offset by an increase among those graduating from restrictive state medical schools.³⁶ Evidence on whether the number of practicing ob-gyns serving patients in restrictive states changed following *Dobbs* is also mixed.^{37,38}

How the National Residency Match Functions

Each year, the National Resident Matching Program (NRMP) uses the stated preferences of applicants and programs to create “matches” with the goal of clearing the labor market for new physicians. The algorithm works in 4 distinct steps: (1) programs register quotas for new residents with the NRMP, (2) applicants and programs rank each other in order of most to least preferred, (3) each applicant’s top choice is revealed, programs select their top choice, and the rest are rejected (4) this process is repeated with the rejected pool of candidates until no program-applicant pair is willing to trade with another. The fraction of a program’s quota that they fill with new residents is known as the “fill rate”. Failure to obtain high fill rates can carry long-term consequences for programs, both financially and in reputational damage that can have knock-on effects for filling residency slots in the future, as unfilled positions can be a red flag for applicants.³⁹

How Programs May Respond to Restrictive Abortion Policy

The Match algorithm was designed to eliminate the benefit of strategically misrepresenting preferences.^{40,41,42,43} For example, if an applicant believed they would be more likely to get a higher ranking from a program located in a restrictive state, but preferred a non-restrictive state program, ranking the restrictive state program higher

would not improve their overall odds of matching and would lower their chances of matching to the more preferred program. Programs, however, can benefit from strategically misrepresenting their preferences, and may select on less-preferred applicants to avoid low fill rates.

Residency positions are heavily subsidized by Medicare, and while each hospital receives effectively the same funding per-position, the maximum number of positions that are eligible for funding is hospital-specific and has remained largely unchanged since 1997.⁴⁴ Rarely, hospitals will self-fund additional positions but the Accreditation Council for Graduate Medical Education (ACGME) ultimately sets a ceiling on program quotas based on case volume, mix, and capacity, that effectively determines the maximum number of residents training in each specialty within a hospital. It is possible that program directors, facing an uncertain applicant pool, may choose to reduce quotas, thus having fewer positions to fill and making it easier to attain high fill rates. As such, programs can strategically drive higher fill rates by reducing quotas or by relying on less-preferred applicants to fill them.

It is unlikely that programs can raise wages to attract applicants, as there is no individual salary negotiation in the Match, and the funding mechanism for residency positions largely suppresses wage competition.⁴⁵ While there are small program-level differences in first-year resident pay across the US, cost of living explains most of this variation, even after accounting for the presence of a union.^{46,47,48}

Methods

Data

Data Sources

Data came from the 2019-2024 annual reports on the Match published by the NRMP, supplemented with resident demographic data from the American Medical Association (AMA) Masterfile. The NRMP data contain information on each program's quota size, and the number of positions filled each year. The AMA data include information on each resident's degree type (MD, DO), sex, birthplace, birth year, the location of their medical training, the year they received their medical degree, and whether their medical training was completed outside of the US (i.e., international medical graduate, or "IMG" status).

Analytic Sample Construction

Within the rank-ordering system, each program is associated with two unique identifiers – one code that corresponds to the NRMP, and a separate institution code corresponding to the ACGME, which when combined with a medical specialty, uniquely identifies each program. The NRMP include their own identifier in their annual reports, but not the ACGME identifier, which are the only program identifiers available in the AMA data. To link the AMA and NRMP data, I constructed a crosswalk of their program identifiers using the NRMP website's program search page – the only public source where these identifiers are co-listed. To be consistent with the NRMP data, resident-level information in the AMA data was aggregated to the program-year level.

Measures

-Outcomes

Our primary outcome was the program fill rate, which we calculated by dividing each program's total matches by its quota for new residents. Program quota was also used as an outcome measure, to assess year-to-year changes in program size that are not captured by fill rates.

-Independent Variables

Several program characteristics were measured to characterize the share of each program's filled positions that were: female, IMGs, osteopathic physicians (DOs), graduates of a medical school located in an abortion restrictive state, and those that were born in an abortion restrictive state, in each year.

-Definition of an Abortion Restriction

We classified abortion policies into one of three categories based on the gestational age after which abortion is prohibited, including: (1) up to 6-weeks' gestation, (2) between 7-18 weeks' gestation, and (3) after 18 weeks' gestation or no restriction. Indicators of state abortion restriction status were constructed based on these three categories and whether the restriction had been adopted by March of a given year. March is when applicants and programs finalize their rank-order listings and thus the last month in which the legal status of abortion can factor into match preferences. We also flagged several states where the legal status of abortion was ambiguous due to ongoing litigation, which were excluded from all analyses. In the year prior to *Dobbs*, Texas began restricting abortion up to 6-weeks' gestation via a system of civil penalties that circumvented *Roe* protections. Because of its proximity and similarity to policies

enabled by the *Dobbs* opinion,⁴⁹ we allowed Texas' treatment to begin with the adoption of this restriction in 2021. This decision was scrutinized in sensitivity testing described later on. Figure 1 displays these measures, showing the legal status of abortion across states in each post-*Dobbs* Match year, in addition to Texas's pre-*Dobbs* restriction.

Policy Exposure

Restrictions up to 6-weeks' gestation are widely considered complete bans on abortion, as these policies provide too narrow of a window to reasonably obtain one.⁵⁰ As such, we considered states with abortion restrictions up to 6-weeks' gestation "restrictive", and those where abortion is legal beyond at least 18 weeks' gestation as "non-restrictive", following classifications established by the Guttmacher Institute.⁵¹ We examine the role of less-restrictive policies between 7 and 18 weeks' gestation in sensitivity analyses described later on.

Statistical Design

Leveraging variation in state abortion policies related to the *Dobbs* opinion, we estimated stacked difference-in-differences models comparing match outcomes at programs located in abortion-restrictive states to those in non-restrictive states, before versus after restriction adoption. Stacked models were used to account for staggered adoption of abortion policies across states, as standard two-way fixed-effects estimators can be biased in settings with heterogeneous treatment timing.⁵² To model these three outcomes, stacked difference-in-differences models took one of the following forms:

Impact of restriction adoption on program fill rates (supply of residents)

$$(1) \log(m_{ptc}) = \beta Treat_{ptc} + \pi_{pc} + \log(Quota_{ptc})$$

$$Matches_{ptc} \sim \text{Poisson}(m_{ptc})$$

Impact of restriction adoption on program quota size (demand for residents)

$$(2) \log(q_{itc}) = \beta Treat_{itc} + \psi_{ic} + \tau_{tc}$$

$$Quota_{itc} \sim \text{Poisson}(q_{itc}),$$

$$Weights: w_{ic} = \mathbb{E}[Quota_{itc} | t < 0]$$

Impact of restriction adoption on program sex composition

$$(3) Female_{ptc} = \beta Treat_{ptc} + \pi_{pc} + \tau_{tc} + \epsilon_{ptc}$$

Where p is a program (the combination of an institution, i , and a medical specialty), t is time in years relative to restriction adoption, and c is a treatment cohort index denoting groups of states by year of restriction adoption.⁵³ $Treat$ is a treatment indicator equal to one in states with an active, 0-6 weeks' gestation abortion restriction (restrictive states), and equal to 0 in states with where abortion is legal beyond at least 18 weeks' gestation (non-restrictive states). $Matches$ is a count of program matches, $Quota$ is a program's pre-registered quota for new residents, and $Female$ is the proportion of female residents that matched to a program. π is a program-fixed effect, ψ is an institution fixed-effect and τ is a year fixed-effect. ϵ is an idiosyncratic error term. In addition to program-level estimates, we also reported overall institution-level changes in average program fill rates and cohort sex composition. Model (2) was estimated solely at the institution-level, given quota allocations are determined at the institution level.

Poisson regression was used to estimate models (1) and (2) for the count structure of the outcome variables. Ordinary least squares regression was used to estimate model (3). (1) captures changes in resident supply by modelling total matches with program quota as an offset to estimate changes in matches per available slot, i.e., the program

fill rate. We took this approach instead of modeling program fill rates directly, as doing so could misattribute changes in program quota sizes to changes in resident supply. As such, (2) was specified to capture changes in program demand by separately modeling program quotas. This model was weighted by the program's average pre-treatment quota, to account for differences in baseline program size. Models were estimated with a washout period for the 2020-2021 match cycle to account for responses to the COVID-19 pandemic,⁵⁴ and standard errors were clustered at the program level.

Assessments of the Parallel Trends Assumption

The difference-in-differences design relies principally on the assumption of parallel trends: that absent the adoption of a restrictive abortion policy, outcomes in states with and without such policies would have evolved similarly over time. We generated evidence on the validity of this assumption by computing and plotting mean trends of all three outcomes (program fill rates, quotas, and sex composition) across the study period, stratified by state abortion restriction status. We also estimated event study analogs of these trends, via the stacked difference-in-differences model specifications described above.

Covariate Adjustment

To investigate key sources of conditional variation, we estimated two covariate adjusted models in addition to our main, unadjusted models.^{55,56} First, we evaluated the influence of underlying residency pipelines on our main findings by estimating model (1) adjusted for the share of a program's residents who graduated from or were born in a restrictive

state and the share who remained in the same state for both medical school and residency. Second, we also estimated this model adjusted for female resident share, to account for bias that may arise because of underlying correlation between resident sex and choice of specialty. Third, we estimated model (2) adjusted for program share of international medical graduates (IMG), who have historically been less competitive than US trained applicants in the Match, to account for potential increased reliance on these applicants to fill positions as an alternative to reducing overall quotas. Finally, we further assessed this relationship by estimating this model with interaction term for IMG share and plotting estimated treatment effects on quota against share of IMGs.

Robustness Testing

First, to scrutinize our choice to estimate models with a washout period for the COVID-19 pandemic, we re-estimated models without the washout period. Second, we assessed restriction severity by relaxing our definition of an abortion restriction from states with 6-week gestation limits to those where abortion is legal up to 18 weeks' gestation. Third, we excluded Texas from analyses to assess our models' sensitivity to its size compared to other restrictive states, as well as its adoption of a pre-*Dobbs* abortion ban. Fourth, we tested sensitivity to estimator selection by re-estimating models (1) and (2) via ordinary least squares (OLS). Finally, we examined the influence of our stacked modeling approach by estimating analogous two-way fixed-effects models using "trigger" abortion laws as a source of identifying variation, which are anticipatory bans on abortion that became effective simultaneous upon the overturning

of Roe. There were 13 states (AK, ID, KY, LA, MS, MO, ND, OK, SD, TN, TX, UT, WY) that adopted such laws.

Results

Descriptive Statistics

Before any state adopted an abortion restriction, average residency training program characteristics were similar across states (Table 1). The pre-restriction proportion of female residents in restrictive state training programs was 0.48, the proportion of IMGs was 0.24, and the proportion of DOs was 0.12, all of which deviated less than 0.02 from non-restrictive state training programs over the same time period. This pattern was consistent across the distribution of specialties in restrictive and non-restrictive states (primary care: 23.0% non-restrictive, 24.0 restrictive; internal medicine: 11% across both groups; emergency medicine (5.7% non-restrictive, 4.7% restrictive), obstetrics & gynecology (6.3% non-restrictive, 6.6% restrictive); and pediatrics (4.4% non-restrictive, 5.1% restrictive). These distributions were broadly stable pre-post restriction adoption.

Unadjusted Average Trends

Program fill rates trended slightly downward among both the restrictive and non-restrictive states in the five years leading up to restriction adoption: the average non-restrictive state program had fill rates of 95.0%, 95.2%, 94.9%, 94.8%, and 94.8%, and the average restrictive state program had fill rates of 95.3%, 95.0%, 93.8%, 93.5%, and 93.8%. In the three years following restriction adoption, non-restrictive state program fill rates increased slightly while restrictive state program fill rates decreased: the average non-restrictive state program had fill rates of 95.2%, 95.5%, and 95.5%, while the average restrictive state program had fill rates of 94.3%, 93.7%, 93.3% (Figure 2).

Overall, the median state average program fill rate grew from 95.9% to 96.3% among non-restrictive states, and fell from 94.6% to 92.9% among restrictive states, pre-post restriction adoption (Figure 2).

Program sex composition followed a similar pattern, with the share of female residents evolving similarly among restrictive and non-restrictive state programs before diverging post-restriction. In the five years pre-restriction adoption, female residents accounted for 47.7%, 47.4%, 48.0%, 48.9% and 49.8% of the average non-restrictive state program's matches, and 44.3%, 44.4%, 45.3%, 47.3%, and 49.0% of the average restrictive state program's matches. Across the three years post-restriction adoption, female residents accounted for 51.0%, 51.3% and 52.3% of the average non-restrictive state program's matches, and 49.6%, 48.5%, and 47.9% of the average restrictive state program's matches (Figure 3).

Event Studies

In event studies of program fill rates, estimated differences between the restrictive and non-restrictive states were statistically insignificant in the pre-restriction period, but diverged in the post-restriction period (Figure 4). Compared to non-restrictive states the year prior to restriction adoption, restrictive states differed in average program fill rates by an insignificant -0.5% (95% CI: -3.0%, 2.1%) five years prior, 0.9% (95% CI: -0.8%, 2.7%) four years prior, -0.8% (95% CI: -2.4%, 0.8%) three years prior, and -0.5% (95% CI: -1.9%, 0.9%) two years prior to adoption. Compared to the same time point,

estimated differences in restrictive and non-restrictive state program fill rates following restriction adoption were an insignificant 0.0% (95% CI: -1.7%, 1.7%) in the first year, a significant -2.5% (95% CI: -4.3%, -0.7%) in the second year, and an insignificant -3.8% (95% CI: -7.7%, 0.1%) in the third year post-restriction. Within specialties, ob-gyn, primary care, and internal medicine training programs followed similar patterns, with event studies demonstrating common trends in the pre-period and significant decreases in the post-period. For other specialties, there were either no differences in the post-period, or pre-period trends were unparallel. These include psychiatry, emergency medicine, hospital-based specialties, pediatrics, surgical specialties, and other/advanced training programs.

In event studies of program quotas, estimates of overall post-restriction differences between restrictive and non-restrictive state programs were statistically insignificant, but within ob-gyn and hospital-based training programs, differences were largely substantial and significant (Figure 5). Compared to non-restrictive states the year prior to restriction adoption, the average restrictive state ob-gyn program's quota changed by an insignificant -13.2% (95% CI: -80.0%, 53.7%) five years prior, -31.2% (95% CI: -81.5%, 2.7%) four years prior, -13.0% (95% CI: -62.5%, 36.5%) three years prior, and -41.8% (95% CI: -75.8%, -7.7%) two years prior to adoption. Compared to the same time point, restrictive and non-restrictive state ob-gyn program quota sizes differed significantly, by an average -58.3% (95% CI: -83.2%, -33.5%) in the first year, -52.8% (95% CI: -74.8%, -30.8%) in the second year, and -47.7% (95% CI: -77.6%, -17.8%) in the third year post-restriction. Estimates of hospital-based training programs' quotas were similar in trend

and magnitude. For other specialties, there were either no differences in the post-period, or pre-period trends were unparallel. These include psychiatry, primary care, internal medicine, emergency medicine, pediatrics, surgical specialties, and other/advanced training programs.

Event studies of program sex composition showed that the post-restriction share of female residents fell among restrictive compared to non-restrictive state programs, but only among programs whose pre-restriction fill rates were lower than the sample average (Figure 6). Among these programs, there were no differences in average female resident share across the pre-restriction period: compared to one year pre-restriction, the average difference in female residents was 6.6% (95% CI: -5.8%, 19.1%) five years prior, -0.1% (95% CI: -7.0%, 6.8%) four years prior, 2.3% (95% CI: 1.6%, -2.6%) three years prior, and 1.6% (95% CI: 1.6%, -2.6%) two years prior to restriction adoption. Compared to the same time point, the average difference in female resident share following restriction adoption was -1.7% (95% CI: -5.5%, 2.2%) one year, -7.8% (95% CI: -15.6%, 0.0%) two years, and -25.2% (95% CI: -37.6%, -12.8%) three years post-restriction. Among programs with above average pre-restriction fill rates, average differences in female resident share were similar in magnitude and trend, but estimates were insignificant.

Difference-in-Differences Estimates

Following the adoption of an abortion restriction, fill rates fell by an average -1.2% (95% CI: -2.4%, -0.1%) among restrictive state programs compared to those in non-restrictive

states (Figure 10). Among specialties whose event studies plausibly supported the assumption of parallel trends, estimates were similar: ob-gyn -1.2% (95% CI: -2.1%, -0.4%), primary care -1.8% (95% CI: -4.1%, 0.4%), internal medicine -1.9% (95% CI: -3.3, -0.5%).

Compared to non-restrictive states, ob-gyn program quotas in restrictive states shrank by an average -2.8 slots (95% CI: -3.3, -2.3) and hospital-based programs by an average of -3.1 slots (95% CI: -3.8, -2.4) relative to pre-restriction averages (Figure 9). After conditioning on program IMG share, magnitudes grew overall: ob-gyn -3.2 slots (95% CI: -3.8, -2.6), hospital-based -3.0 slots (95% CI: -3.9, -2.2; Figure 9). Interacted models showed that IMG share may have mediated the relationship between abortion restriction adoption and quota size (Figure 10). Lower IMG share was associated with restrictive state programs reducing quotas – with 0% IMG share corresponding to an estimated reduction of -2.1 slots (95% CI: -3.5, -7.3) – while higher IMG share corresponded to maintained or even increased quota sizes – 50% IMG share was associated with an estimated reduction of -0.5 slots (95% CI: -2.3, 1.4), and 100% IMG share with an estimated increase of 1.6 slots (95% CI: -2.1, 5.3).

Among programs with average fill rates below the pre-restriction sample mean (95.0%), female resident share fell by 5.7 percentage points (95% CI: -9.6 ppt., -2.0 ppt) in restrictive states compared to programs in non-restrictive states, after versus before restriction adoption (Figure 11). There were no such changes among programs with fill

rates above the pre-restriction sample mean (Estimate: -0.8 ppt., 95% CI: -3.4 ppt., 1.7 ppt.), nor overall (Estimate: -0.6 ppt., 95% CI: -2.6 ppt., 1.5 ppt.).

Results of Robustness Testing

Adjusting for female resident share did not meaningfully influence point estimates of models assessing impacts on program fill rates: ob-gyn -2.0% (-3.5%, -0.4%), primary care -2.7% (-5.5%, 0.2%), -1.8% internal medicine (95% CI: -3.4%, -0.1%). After conditioning on prior residence in a restrictive state, magnitudes grew: overall -2.1% (95% CI: -3.9%, -0.2%), ob-gyn -3.0% (-4.9%, -1.1%), primary care -2.3% (-4.9%, 0.4%), -2.6% internal medicine (95% CI: -4.4%, -0.9%). Estimates of abortion restriction adoption on program fill rates via models that did not exclude a washout period for the COVID-19 pandemic corresponding to the 2020-2021 match cycle were noisier but largely consistent with main findings (Overall: -0.9%, 95% CI: [-2.0%, 0.2%], ob-gyn: -0.8%, 95% CI: [-1.8%, 0.3%], primary care: -1.4%, 95% CI: [-3.6%, 0.8%], internal medicine: -1.9%, 95% CI: [-3.3%, -0.4%]). In models excluding Texas from the analysis, point estimates increased slightly relative to main findings (Overall: -1.3%, 95% CI: [-2.8%, 0.2%], ob-gyn: -1.5%, 95% CI: [-2.7%, -0.2%], primary care: -1.8%, 95% CI: [-4.6%, 1.0%], internal medicine: -2.3%, 95% CI: [-4.2%, -0.4%]). Conversely, point estimates diminished slightly in models where we relaxed our definition of an abortion restriction to incorporate limitations of up to 18 weeks' gestation (Overall: -1.0%, 95% CI: [-2.2%, 0.1%], ob-gyn: -1.0%, 95% CI: [-1.9%, -0.2%], primary care: -1.6%, 95% CI:

[-3.9%, 0.6%], internal medicine: -1.7%, 95% CI: [-3.1%, -0.4%]). Models estimated via OLS were also consistent with main findings estimated via poisson regression.

Patterns were similar in analyses of program quotas, including models without a COVID-19 washout period (ob-gyn: -2.9 slots, 95% CI: [-3.5, -2.3], hospital-based medicine: -3.2, 95% CI: [-4.0, -2.5]), excluding Texas (ob-gyn: -2.4 slots, 95% CI: [-2.9, -2.0], hospital-based medicine: -2.5, 95% CI: [-3.2, -1.7]), and those using the relaxed definition of an abortion restriction of up to 18 weeks' gestation (ob-gyn: -2.8 slots, 95% CI: [-3.3, -2.3], hospital-based medicine: -2.9, 95% CI: [-3.6, -2.3]). Additionally, estimations via OLS were consistent with main analyses that used poisson regression for estimation, as were two-way fixed effects models that used states with trigger abortion bans pre-post *Dobbs* as a source of identifying variation.

Discussion

In this analysis of national resident match data, we found that states adopting restrictive abortion laws enabled by the 2022 *Dobbs v. Jackson* Supreme Court decision resulted in fewer medical graduates matching to residency programs located in restrictive states compared to those in states where abortion remained legal. These effects were primarily concentrated among maternal and reproductive healthcare specialties, such as obstetrics and gynecology, internal medicine, and primary care training programs. Many affected programs matched with fewer female residents, and either reduced their quotas for new residents or relied more heavily on less-competitive applicants to fill them.

Figure 1. Legal Status of Abortion Across States During the Match in Each Year

Since the First Observed Restriction was Adopted

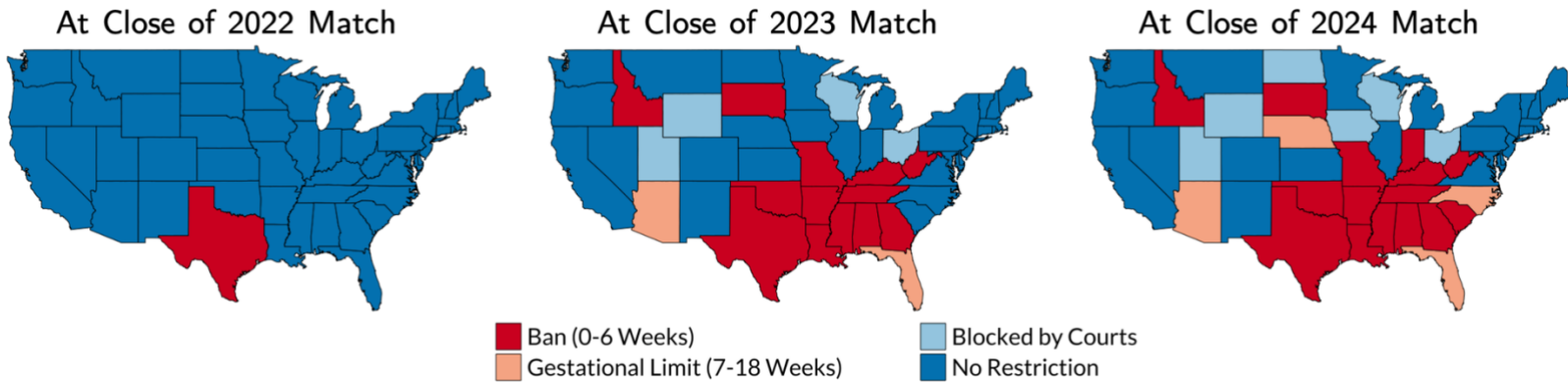


Table 1. Descriptive Statistics of Programs Located in Restrictive and Non-Restrictive States in the Years Prior to and Following Restriction Adoption

Timing	Pre-Restrictions* (2019-2021)			Post-Restrictions (2022-2024)		
State Abortion Policy	Non-Restrictive	Restrictive		Non-Restrictive	Restrictive	
Program-Years ^a	N = 7,599	N = 2,751		N = 7,968	N = 2,772	
	Mean/Proportion (SD); n (%)		p-value ⁺	Mean/Proportion (SD); n (%)		p-value
Cohort Characteristics						
Graduated from Restrictive State Medical School	0.11 (0.19)	0.51 (0.33)	<0.001	0.14 (0.20)	0.48 (0.30)	<0.001
Born in Restrictive State	0.06 (0.15)	0.26 (0.30)	<0.001	0.10 (0.17)	0.36 (0.27)	<0.001
US Born	0.74 (0.31)	0.78 (0.29)	<0.001	0.77 (0.26)	0.81 (0.24)	<0.001
Female	0.48 (0.33)	0.46 (0.33)	0.13	0.50 (0.29)	0.49 (0.29)	>0.9
International Medical Graduates (IMG)	0.24 (0.35)	0.21 (0.32)	0.50	0.22 (0.32)	0.19 (0.28)	>0.9
Doctor of Osteopathy (DO)	0.12 (0.24)	0.12 (0.23)	>0.9	0.13 (0.23)	0.13 (0.21)	>0.9
Matches	8.75 (9.84)	8.42 (8.56)	>0.9	11.42 (27.14)	11.34 (21.27)	>0.9
Quota	9.21 (9.95)	8.97 (8.78)	>0.9	12.05 (27.98)	12.12 (22.81)	>0.9
Specialty						
Hospital-Based	1,182 (16.0%)	399 (15.0%)	>0.9	1,212 (15.0%)	399 (14.0%)	>0.9
Emergency	436 (5.7%)	128 (4.7%)	0.60	459 (5.8%)	129 (4.7%)	0.5
Primary	1,730 (23.0%)	666 (24.0%)	>0.9	1,827 (23.0%)	675 (24.0%)	>0.9
Internal Medicine	816 (11.0%)	307 (11.0%)	>0.9	852 (11.0%)	309 (11.0%)	>0.9

Ob-Gyn	480 (6.3%)	181 (6.6%)	>0.9	492 (6.2%)	183 (6.6%)	>0.9
Pediatrics	338 (4.4%)	141 (5.1%)	>0.9	345 (4.3%)	141 (5.1%)	>0.9
Psychiatry	459 (6.0%)	155 (5.6%)	>0.9	495 (6.2%)	156 (5.6%)	>0.9
Surgery	1,596 (21.0%)	559 (20%)	>0.9	1,686 (21.0%)	561 (20.0%)	>0.9
Other/Advanced	321 (4.2%)	137 (5.0%)	>0.9	333 (4.2%)	138 (5.0%)	>0.9
Transition Year	241 (3.2%)	78 (2.8%)	>0.9	267 (3.4%)	81 (2.9%)	>0.9

*: Of the 50 states and Washington DC, restrictive states were those that had an abortion restriction in place at any point following the Dobbs decision

(AK,ID,KY,LA,MS,MO,ND,OK,SD,TN,TX,UT,WY,AL,WV,WI,GA,NB,FL,AZ,IN,SC,NC)

+: Pearson's Chi-squared test; Wilcoxon rank sum test, Bonferroni adjusted

a: A "program" is comprised of an institution and a specialty

Figure 2: Annual Average Program Fill Rates Relative to Restriction Adoption

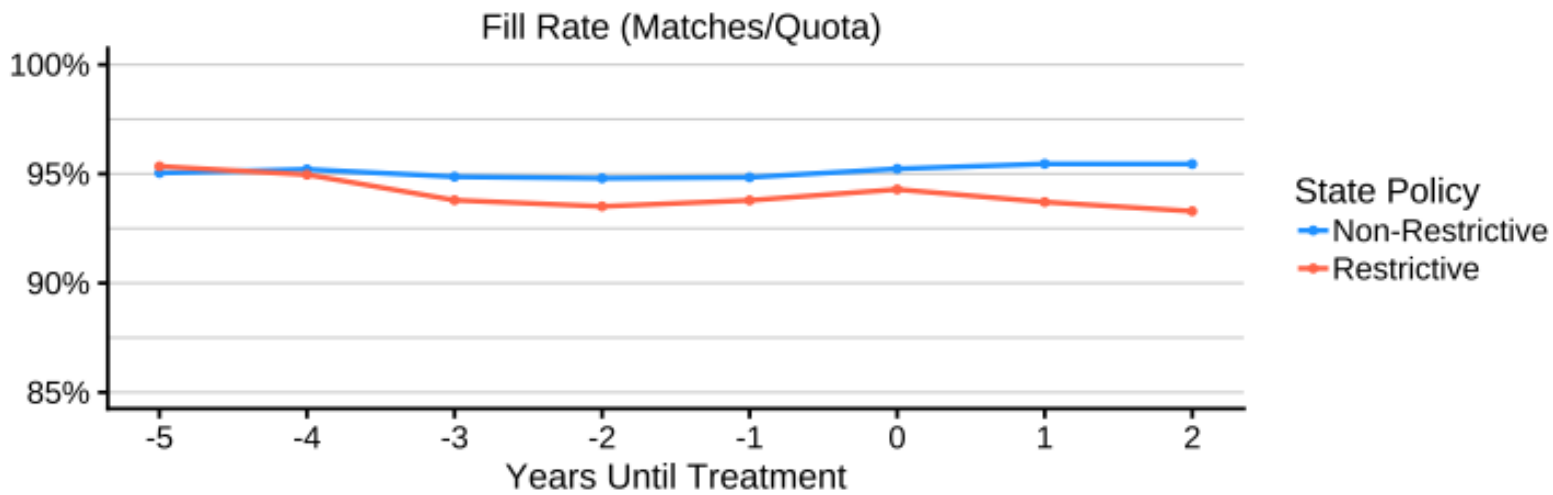


Figure 3: Annual Share of Female Residents in the Average Program, Relative to Restriction Adoption

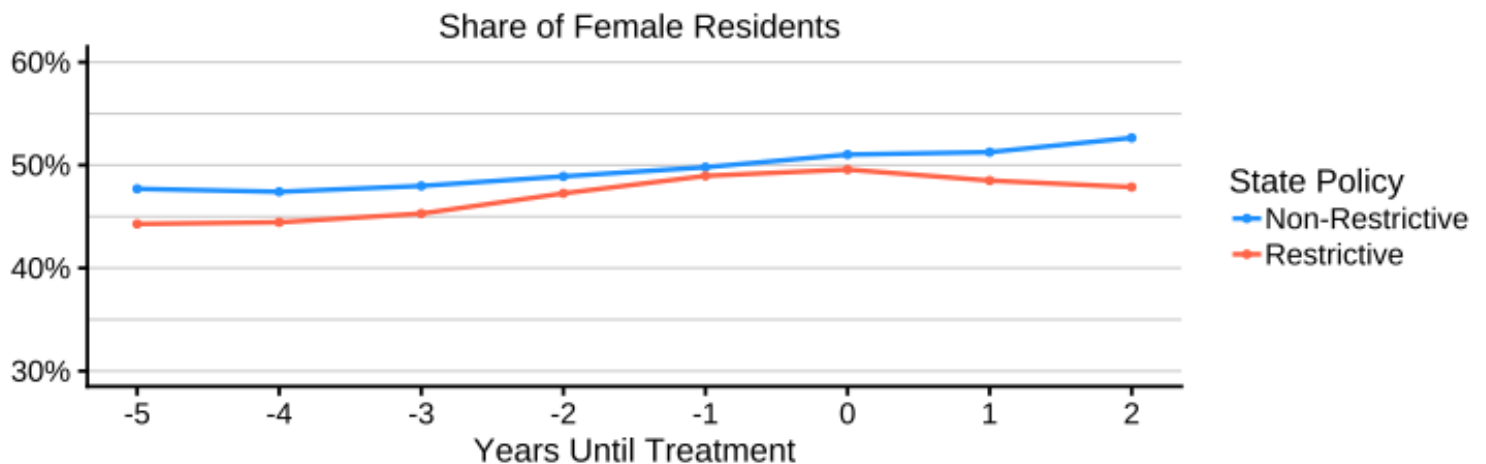


Figure 4: Event Studies of Program Fill Rates by Specialty

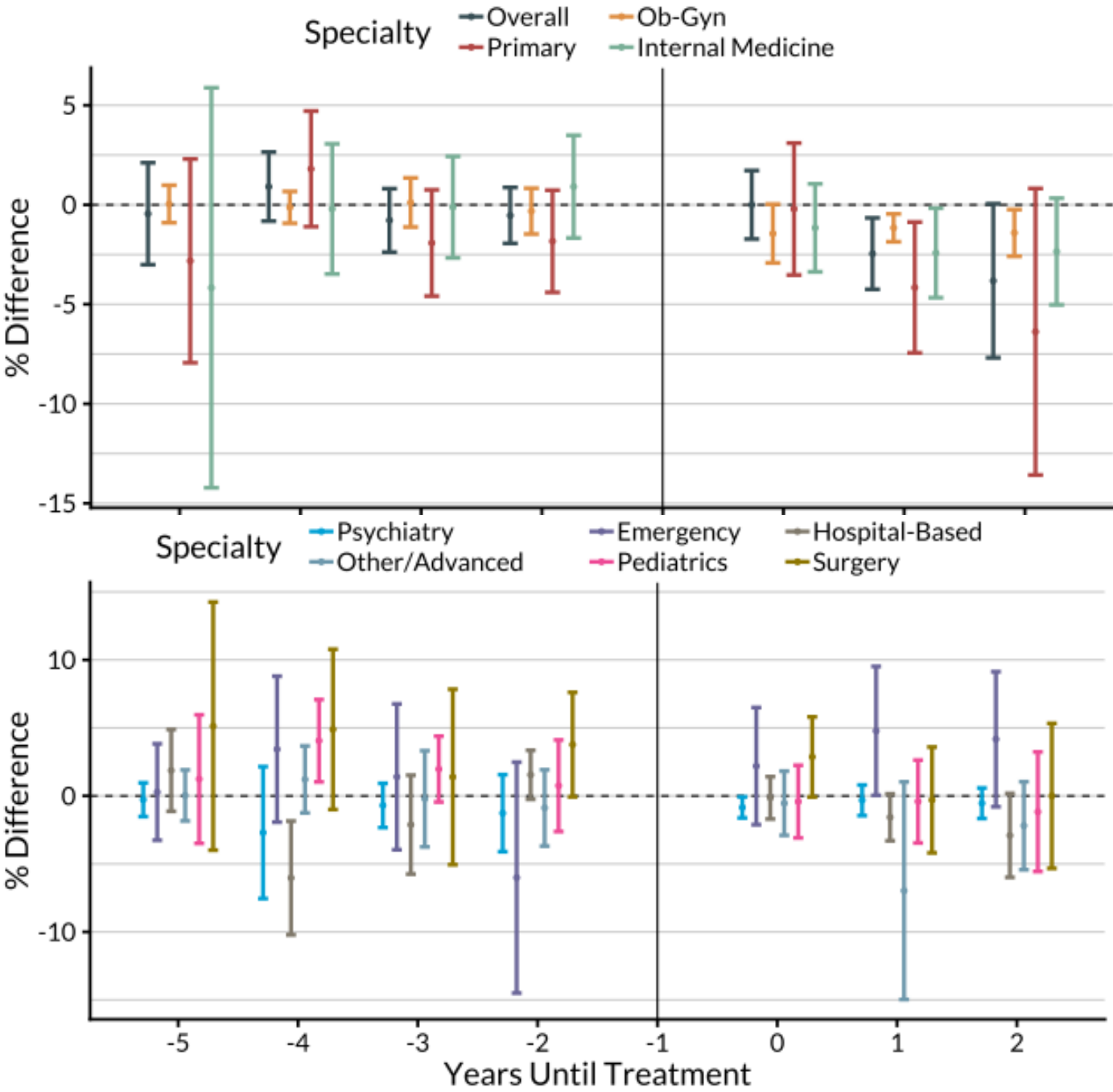


Figure 5: Event Studies of Program Quotas by Specialty

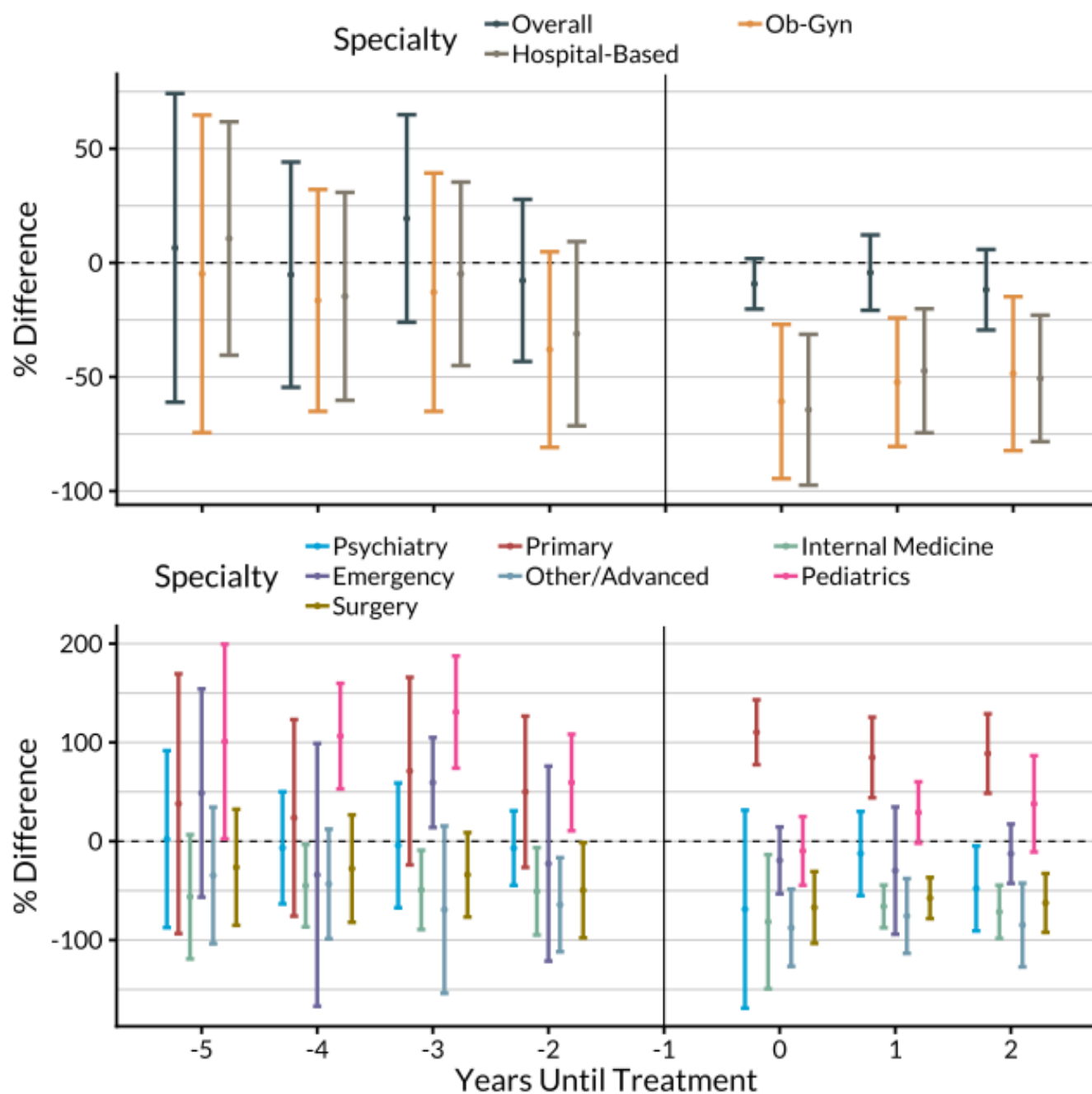


Figure 6: Event Studies of Program Sex Composition, by Pre-Period Fill Rate

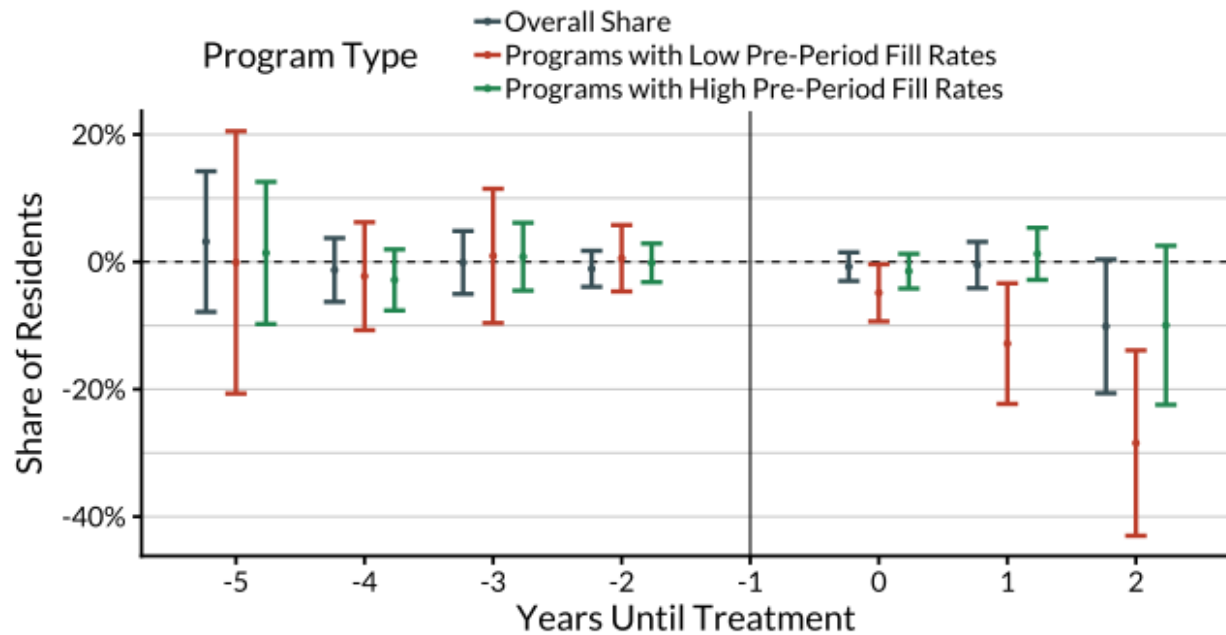


Figure 7: Significant Difference-in-Differences Estimates of Effects on Program Fill Rates by Specialty

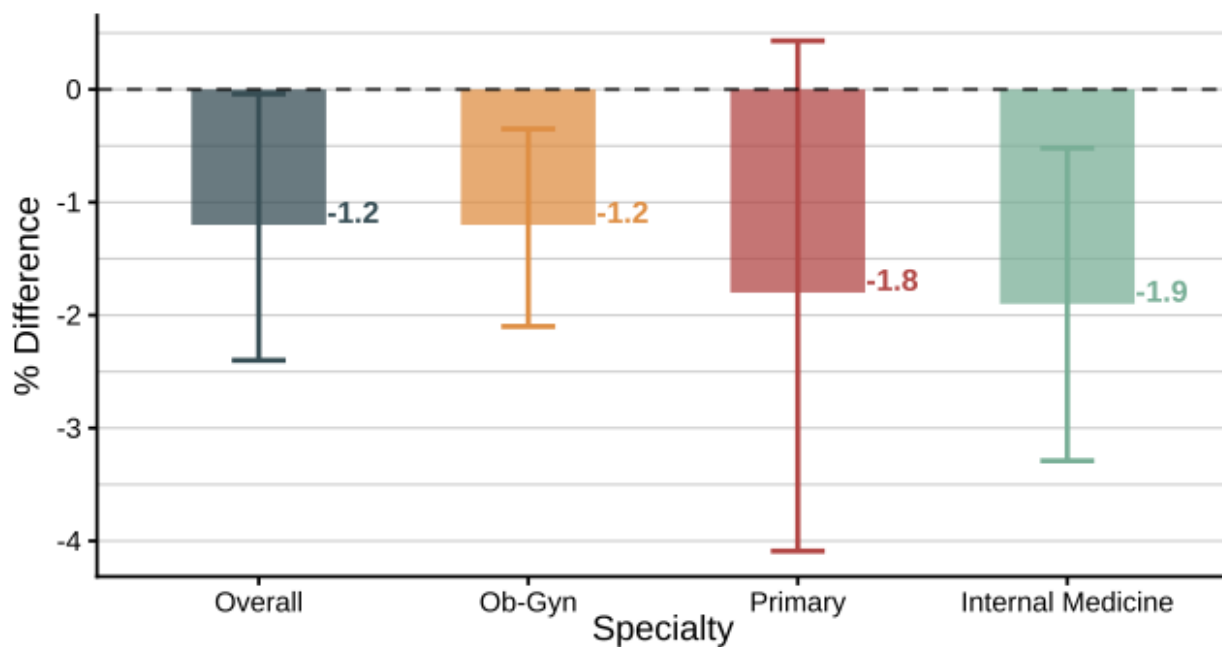


Figure 8: Significant Difference-in-Differences Estimates of Effects on Program Quota by Specialty

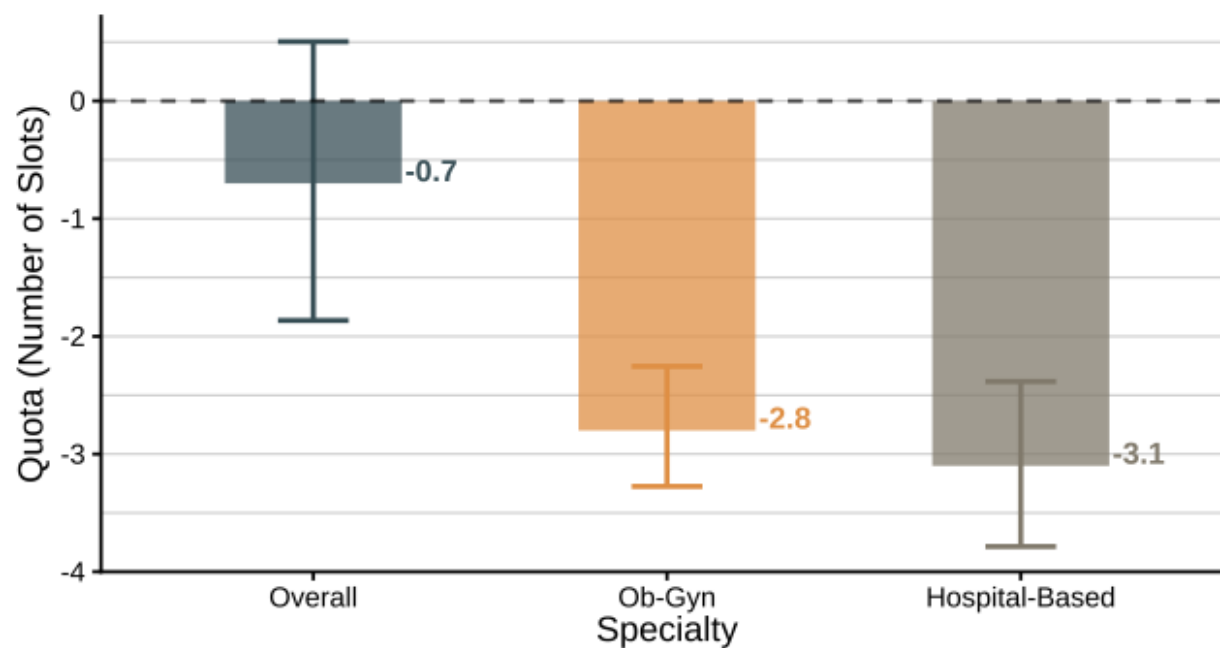
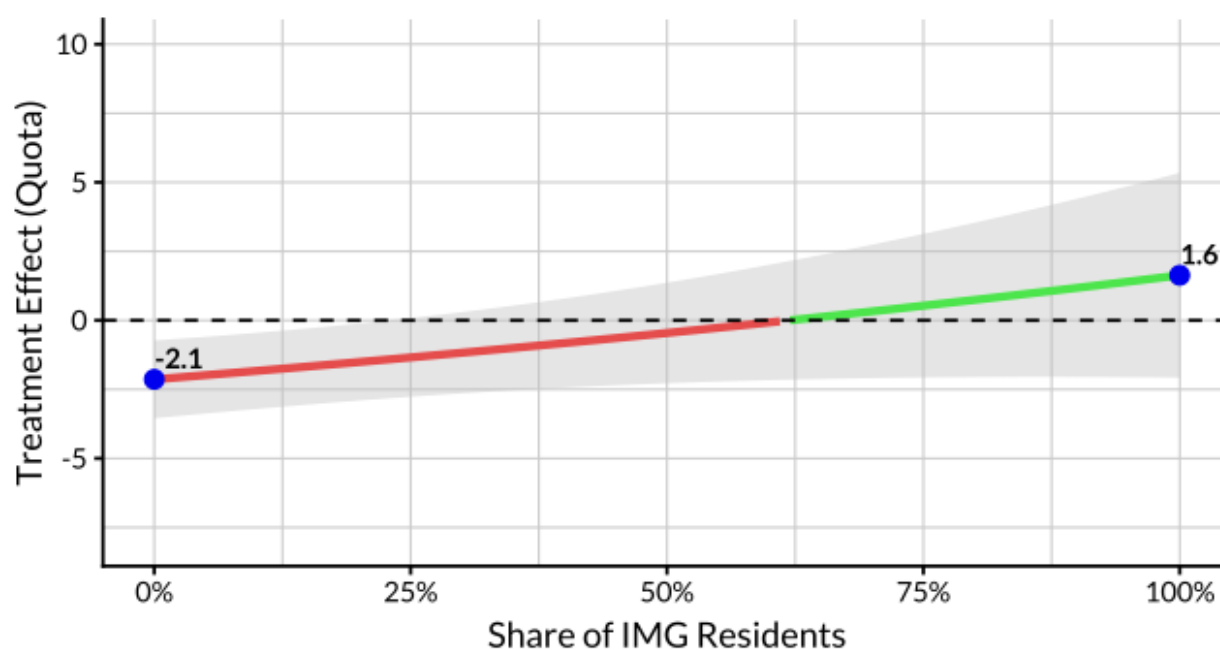
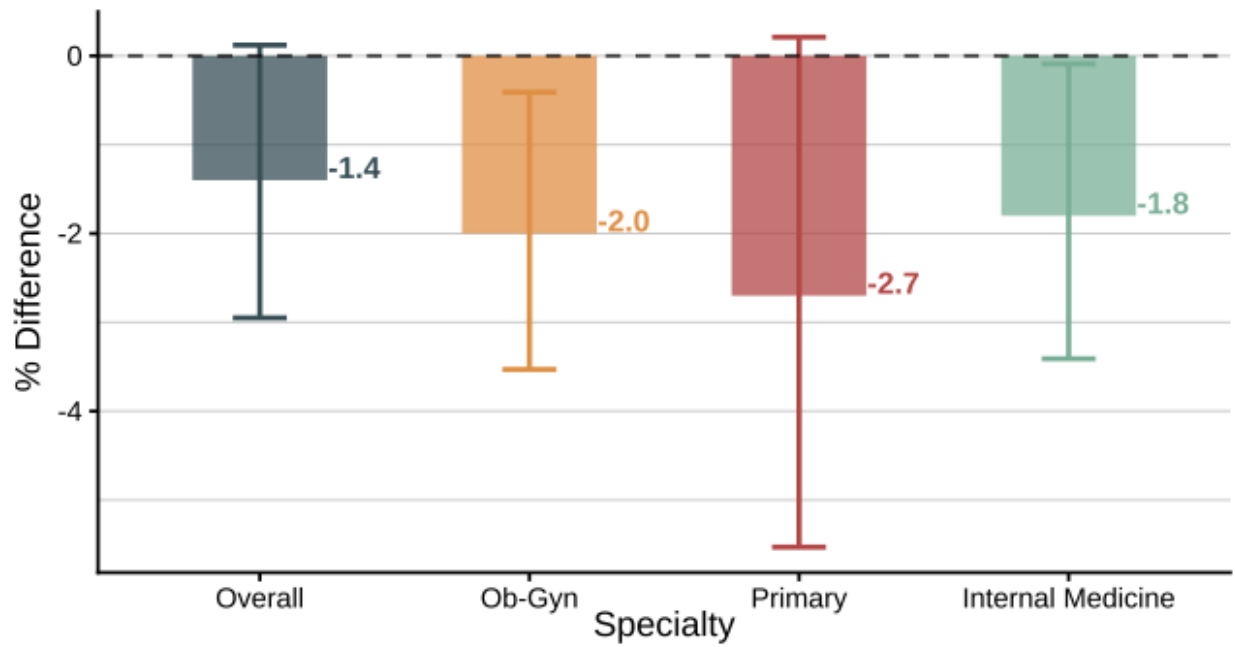


Figure 9: Difference-in-Differences Estimates of Effects on Program Quota by IMG Share

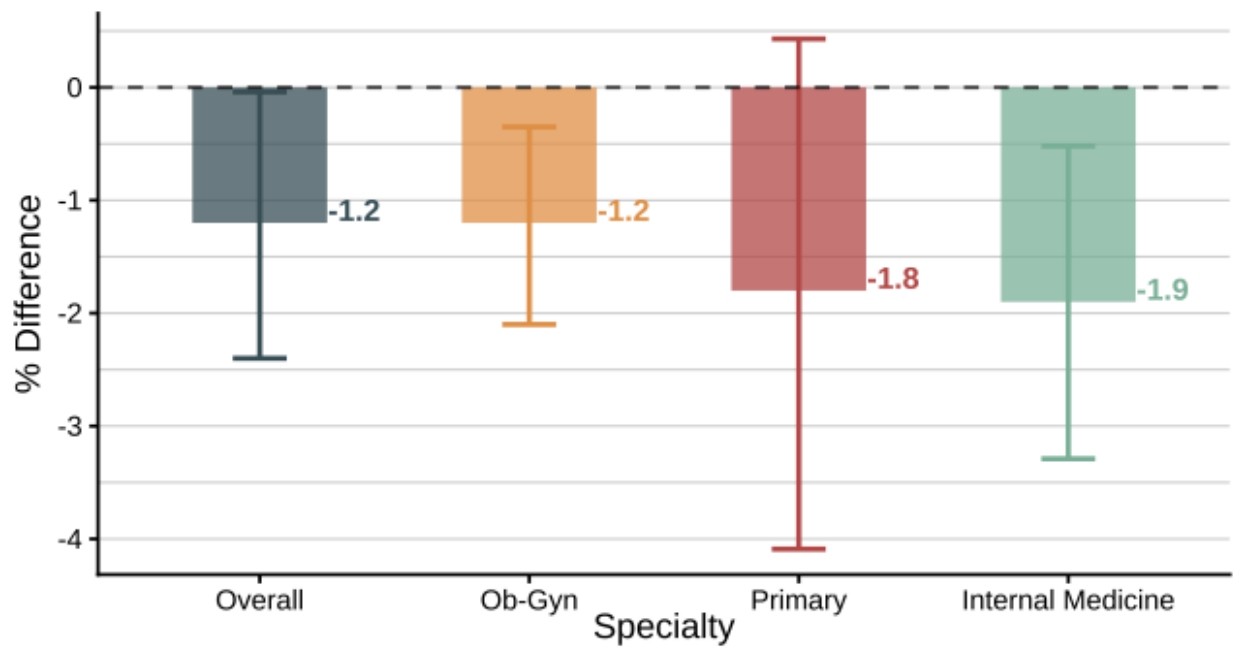


Appendix Tables and Figures

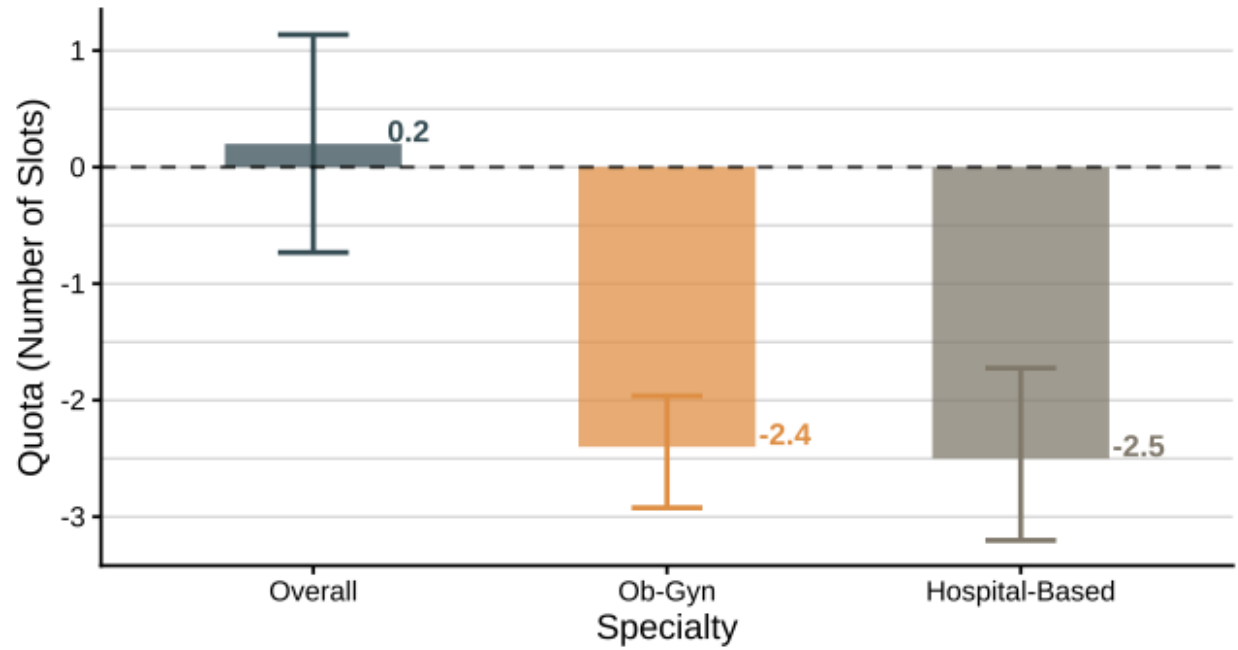
Match coefplot, female adjust



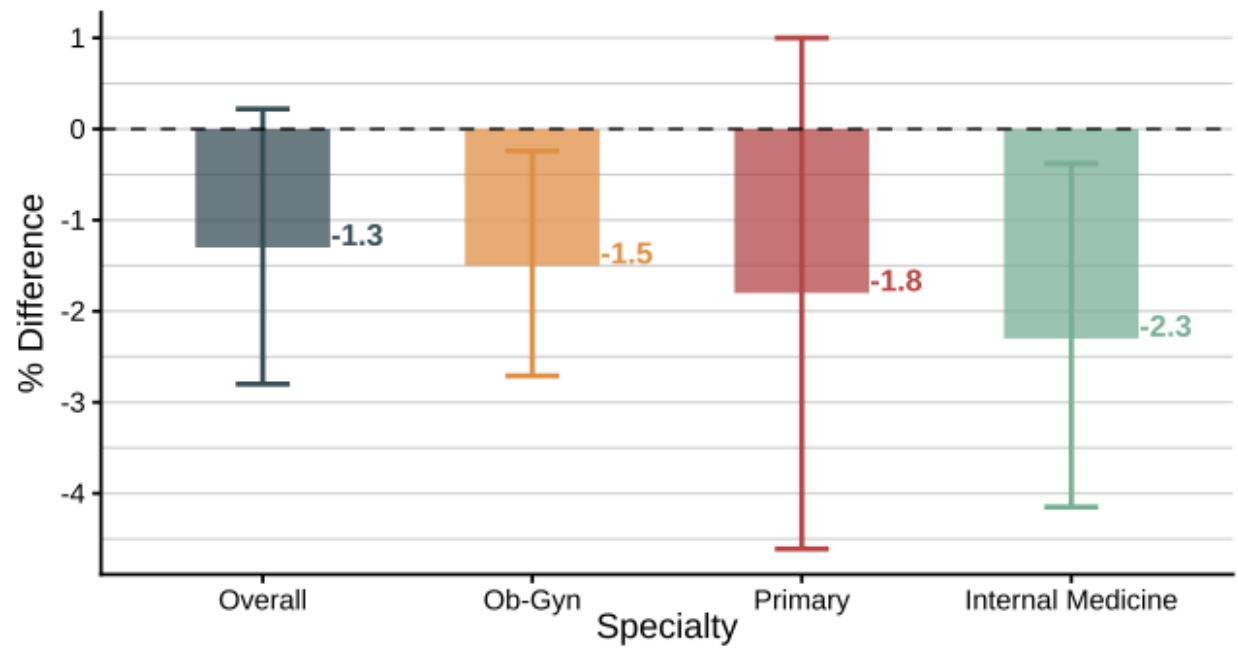
Match coefplot, priorstate adjust



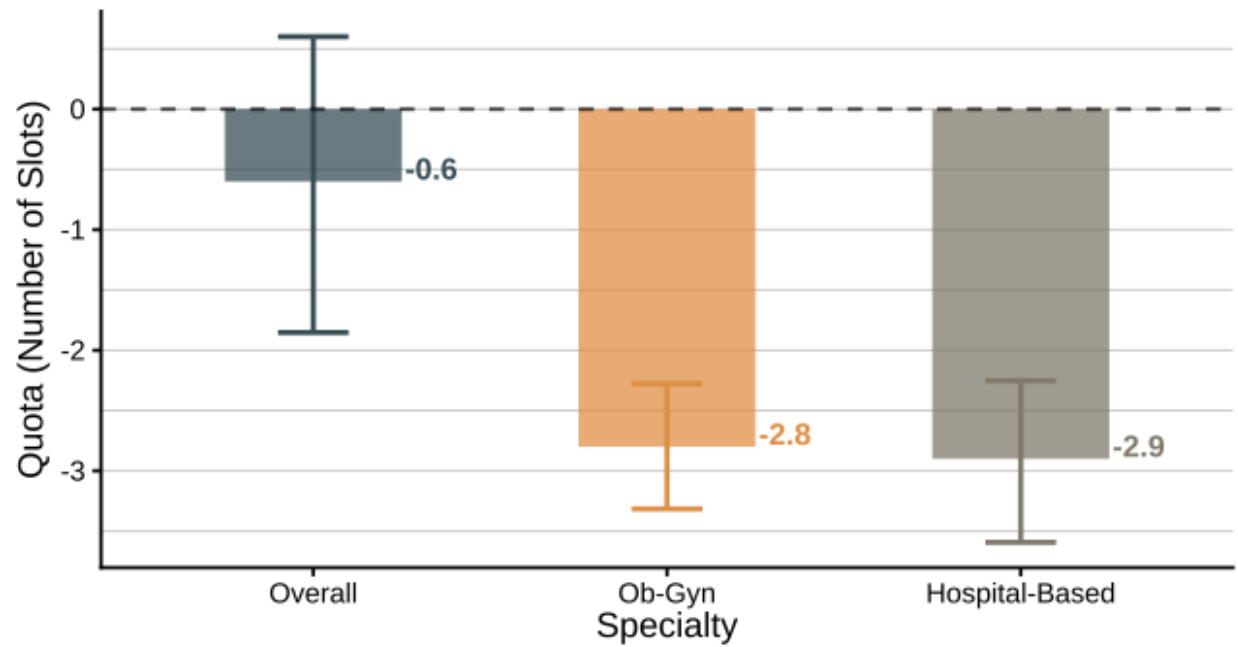
Quota, Omit Texas



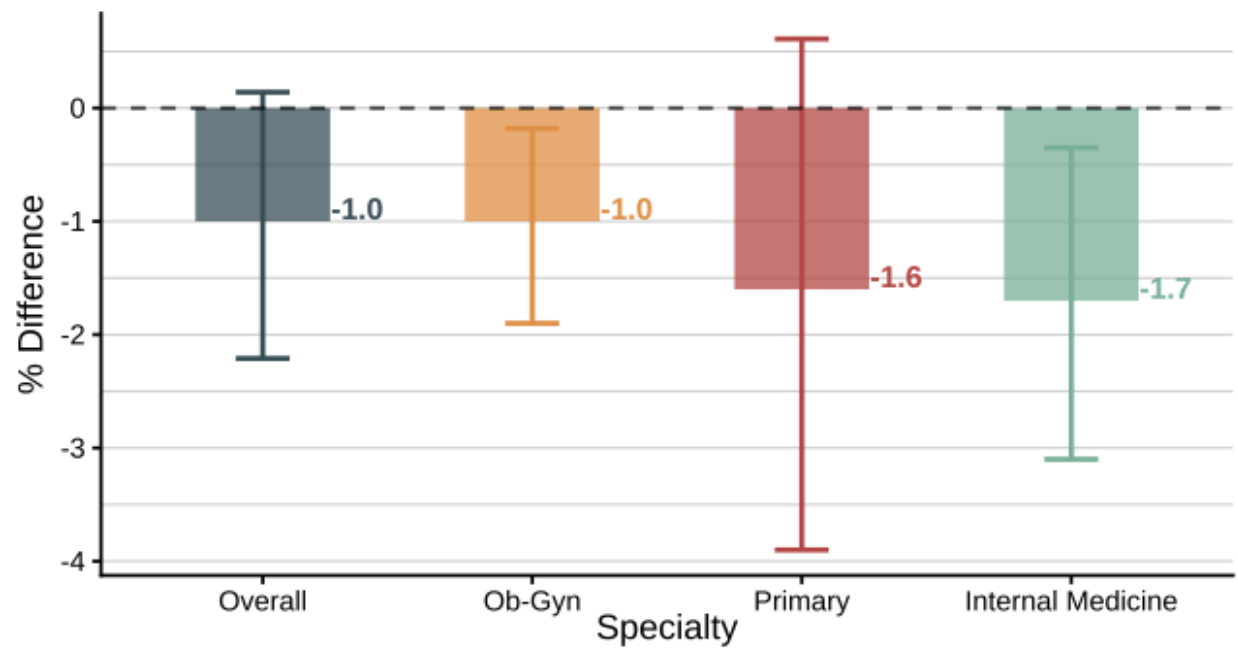
Match, omit texas



Quota, weak restrictions



Match, weak restrictions



Theoretical Model

To understand the decision-making of residency applicants, I consider a model of labor supply at the extensive margin based on the hedonic wage model, a framework for labor market equilibrium based on the decomposition of jobs into features, and subsequently, the utility each feature confers to the worker.⁵⁷ In this setting, a program can be decomposed into a bundle of features, with each resident obtaining some level of utility from those features. In particular, the hedonic model allows us to consider how a change to one such feature, such as preference for geographic location, affects equilibrium.

Each applicant's utility depends on a non-pecuniary bundle of features that define each residency program (number of faculty, geographic location, etc.), and, their stock of health, following the Grossman model of health production.⁵⁸ Programs do not have the ability to negotiate salaries with applicants directly, due to the structure of the centralized matching market. As such, my conceptualization deviates from the canonical hedonic wage model by focusing on non-pecuniary program features, rather than wages. The core results of the model are unaffected by this change.

Defining Utility

Without loss of generality, let the j^{th} individual be a labor force participant such that their utility U_j depends on wages W , the qualities of their job Z_i , $i \in [1, \dots, n]$, and a stock of health H .

$$U_j = (\bar{W}, Z_1, \dots, Z_n, H) \quad (1)$$

For simplicity, it is assumed that all income comes from wages, though the inclusion of non-labor income, savings, and spousal income would not alter the implications of the model. Importantly:

$$\frac{\partial U_j}{\partial H} > 0, \quad \frac{\partial^2 U_j}{\partial H^2} < 0 \quad (2)$$

That is, utility is assumed increasing and concave in health stock. Following insights from Grossman, 1972, individuals have an initial endowment of H , which can be modulated by trading off wages W for health improving consumption either explicitly, like through the purchase of medical care, or implicitly, via working fewer hours (thereby reducing W along the intensive margin¹) to reduce sick time, for example.⁵⁹ That is:

$$\frac{\partial^2 U_j}{\partial H \partial Z_i} > 0 \quad (3)$$

Which indicates that a better working environment, as defined by beneficial job characteristics, increases the stock of health. It is important to note that this treatment of utility and the production of healthy time does not take a life-cycle approach as prescribed by Grossman's model, though for consistency, utility defined in this model can be thought of as being a cross-section, observed at a single point in time.

Letting Z_1 represent the geographic location of a given job, consider a shock $-\epsilon$ to the mental health of a given resident of a restrictive state j' , who is assumed similar to a given non-restrictive state resident j :

$$U_j = (\bar{W}, Z_1, \dots, Z_n, H) < U_{j'} = (\bar{W}, Z_1 - \epsilon, \dots, Z_n, H') \quad (4)$$

¹ Since all individuals are assumed labor force participants in this model, the extensive margin need not be considered.

where since Z_1 is positively related to H , its reduction causes the stock of health to diminish to $H' < H$. Consequently individuals in restrictive states can offset the mental health impacts to H by leaving the restrictive state for a new job in a non-restrictive state, or by improving their other job features $Z_i, i \neq 1$, W to compensate. The next section will focus on this type of offsetting behavior by employing the Hedonic Model in the context of job seeking – in this case, among medical residency applicants.

Overview of the Hedonic Model

In the simplest formulation of the model, we consider only the worker's choice, assuming that the jobs on offer are fixed, heterogenous, and there are more than two, such that there is a non-degenerate choice available to the jobseeker. Job characteristics are also assumed to be continuous, such that the menu of jobs, as defined by those characteristics, are not "lumpy". A worker's utility is defined by a concave function of job characteristics and wages, positive in all arguments. Wages, however, are defined by the market, so utility maximization is constrained by the "hedonic prices" of characteristics, which is to say, the wages as a function of characteristics, negative in all arguments, reflective of the central tradeoff mentioned previously. This wage function is referred to as the "market locus" and in its simplest form is assumed linear, such that the basic constrained optimization procedure can be employed, i.e., maximizing utility subject to a linear budget constraint. As such, and intuitively, the optimal conditions suggest that the vector of the ratio of marginal utilities of characteristics and wages are equivalent to the hedonic prices of the characteristics in equilibrium. To reach this point, we must assume that risk-preferences are known

(this is especially important for the additions that Rosen makes, discussed later), that there is full information, and that there is no “job-lock”, in which workers cannot move freely between jobs. That said, some of the other assumptions made along the way can be relaxed.

It is unlikely that job characteristics are not lumpy; rather, given a menu of different jobs, it is likely that the relationship between wages and characteristics is non-constant. As such, we can relax the assumption of the linear market locus and can either assume its functional form and derive demand, or more commonly, extrapolate a virtual budget constraint from the intercept and the assumption that the market locus is linear in at least one job characteristic. Additionally, we can stop ignoring the firm side and solve for equilibrium given the profit maximizing decisions of firms and the utility maximizing choice of workers.

The lumpiness of the budget constraint comes from the empirical observation that, contrary to what the existence of an optimum suggests, we see many different job characteristics in the market. One explanation, from Rosen, is that workers have different preferences for wages and for characteristics. As such, he allows for such heterogeneity, which yields multiple tangencies. A similar argument can be made for firms that have different iso-profit curves, and thus are heterogeneous in costs, in the sense that the production of different features may be more or less expensive to any given firm. Thus, we can consider equilibrium in the market clearing sense – where demand for jobs with a given bundle of characteristics is equal to the supply – firms

observe the market locus, and provide a bundle according to their costs, workers then choose according to their preference. This results in a set of tangencies between heterogenous workers and firms, which are sorted efficiently, as workers with strong preferences for certain features over wages will pick those jobs with lower wages and a higher amount of those features, and vice versa. Exogenous shocks, thus, will tilt the market locus steeper or flatter, resulting in more or fewer jobs with a given characteristic or characteristics. This can come from either side of the market; if those characteristics become more or less costly to produce, or more or less valuable to workers if there is some shock to preferences, respectively. It should be mentioned that often these features are conceptualized as either “good things” like fringe benefits, or “bad things” like the danger of operating heavy machinery, depending on the author.

Defining the Residency Applicant in the Hedonic Model

I define a residency as a single good comprised of a bundle of features Z_i , $i \in [1, \dots, n]$, not necessarily continuous. The hedonic price function is defined by the relationship between the wage rate and “amounts” of a bundle of given features, Z_i . That is,

$$\bar{W} = f(Z_1, \dots, Z_n) \quad (5)$$

where f is the hedonic price function and is assumed negative in all arguments. The special case of the residency makes it such that wages are negligibly different from one and other, and thus are assumed to differ only by cost of living. As such, I use the notation $\bar{W}$ to denote that wages are homogenous, and to demonstrate that the critical hedonic tradeoffs happen between job features and other job features, not between wages and job features. That is, if Z_i falls, some $Z_j, i \neq j$ must increase to maintain $\bar{W}$.

Recall from (1) that the utility of the j^{th} applicant is a function of the features of the job, is positive in those arguments, and is concave, in that there are diminishing marginal returns from job features and wages.

$$U_j = (\bar{W}, Z_1, \dots, Z_n, H), \text{ and } \frac{\partial U_j}{\partial Z_i} > 0, \frac{\partial U_j}{\partial \bar{W}} > 0 \quad (6)$$

It should be noted that the inclusion of wages here is in a sense degenerate, as programs do not compete on wages. Nonetheless, it is still true that an increase in the wage rate will increase utility, though in this framework, all residents would experience that increase simultaneously, which given the ordinal properties assumed of the utility function, would not alter any applicant's rank-order list.

Using the usual Lagrangian approach, we have the first order conditions (FOC):

$$\frac{\partial U_j}{\partial \bar{W}} + \lambda = 0 \quad (7)$$

$$\frac{\partial U_j}{\partial Z_i} - \lambda f_i = 0 \quad (8)$$

Thus, we have tangencies at:

$$\frac{\partial U_j}{\partial Z_i} f_i^{-1} + \frac{\partial U_j}{\partial \bar{W}} = 0 \quad (9)$$

Defining the Program in the Hedonic Model

To make the discussion of the agents involved complete, let the “surplus”² function of the t^{th} hospital be described by the quantity and quality of healthcare services provided

² The term “surplus” is used to reflect the hospital's tradeoff between the costs of labor, and the quantity and quality of care provided following Newhouse, 1970.

Q , which depends on the number of residency slots filled L , and the prices associated with those services P , less the cost of labor and residency features, accounting for hospital-specific costs with the parameter β , which in this case, largely captures the economies of scale enjoyed by major prestigious teaching institutions.

$$\pi_t = PQ(L) - f(Z_1, \dots, Z_n)L - C(Z_1, \dots, Z_n, \beta) \quad (10)$$

However, it was previously established that wages are homogenous, thus:

$$\pi_t = PQ(L_t) - \bar{W} * L_t - C(Z_1, \dots, Z_n, \beta) \quad (11)$$

Since the hedonic price function is negative in all arguments, but changes in Z_i cannot be compensated with corresponding changes in $\bar{W}$ since $\frac{d\bar{W}}{dZ_i} = 0$, the critical tradeoff is maintained by the fact that $\frac{\partial \pi_t}{\partial Z_i} < 0$, implying that programs must balance their ability to fill residency slots L against the amounts of the features they produce. Moreover, since wages are homogenous, hospitals cannot price discriminate by offering higher wages to more competitive residents. Thus, hospitals face an additional tradeoff between quantity and competitiveness, assuming that more competitive residents are more likely to gain employment at more competitive programs, as measured by higher Z . Let resident competitiveness be represented by X , the proportion of residents above some threshold of competitiveness in the program.

$$\pi'_t = PQ(L_t, X) - (\bar{W} * L_t) - C(Z_1, \dots, Z_n, \beta) \quad (12)$$

Where $\frac{\partial Q(\cdot)}{\partial X} > 0$ in the sense that the quality and quantity of services provided are increasing in the competitiveness of the residents, arising from efficiency. Thus, the profit maximizing conditions are:

$$\frac{\partial \pi'_t}{\partial L_t} \Rightarrow P \frac{\partial Q(X, L_t)}{\partial X_i} \frac{\partial X}{\partial L_t} = \bar{W} \quad (13)$$

So, hospitals must consider the quantity-competitiveness tradeoff with regards to balancing the marginal product of filling a slot with a competitive resident, or hiring multiple residents that are less competitive, all of which is constrained by the conditionally homogenous wage rate they must pay out to all hires, L . In practice, programs will open an additional slot until the marginal product of the additional slot is equal to the wage, though this is tempered by how much increasing the resident pool affects competitiveness, and the marginal product of a competitive resident, as filling many slots with residents that are less competitive will lower surplus. And:

$$\frac{\partial \pi'_t}{\partial Z_i} \Rightarrow -C'(Z_1, \dots, Z_n, \beta) = \bar{W} L_t \quad (14)$$

Which, since the wage rate is fixed, implies that hospitals can only adjust the number of slots they fill via producing more Z to increase profits up to the point where the marginal costs of production equals volume-based wage savings.

Equilibrium in the Hedonic Model

Considering the i^{th} residency as a bundle of features: $R_i = (Z_1, Z_2, \dots, Z_n)$, and the bundle of wages $F_i = (f_1, f_2, \dots, f_n)$ that an applicant would tradeoff for R_i , equilibrium occurs where demand for residency R_i equals its supply. Programs offer residencies with varying features at the wage rate $\bar{W}$, and applicants choose the residency that maximizes their utility. The market locus is therefore carved out by “kissing curves” generated by the set of resident and program specific tangencies (as hospital side heterogeneities may induce non-linearities), which are predicated on the assumptions of

utility maximization, full information, and mobility. I argue that these are plausible in this setting, given residents have little reason to not strive for the program that best meets their preferences, the institutionally managed job market (by the National Residency Match Program) makes full information probable, and residency applicants tend to be more mobile than other job seekers given the low historical rates at which residents match to their medical degree granting institution relative to the rates at which applicants cross state lines for their residencies.

To model the case of an exogenous shock to applicant preference for some Z_i , let ϵ be some strictly positive value such that:

$$U_j = (\bar{W}, Z_1, \dots, Z_n) > U'_j = (\bar{W}, Z_1, Z_i - \epsilon, \dots, Z_n) \quad (15)$$

All else constant, the supply of applicants for programs containing Z_i will fall:

Since wages are homogenous, these programs will have to reduce L , X , or both, to attract applicants. Programs may be able to recoup supply by increasing other $Z_{j, i \neq j}$, however, shown this is a costly endeavor and will reduce L , X , or both. Consequently, the model suggests that a shock of this kind will constrict the market for programs affected by the shock and given the nature of the two-sided matching labor market, we would expect that the market for unaffected programs will experience a proportional increase in supply.

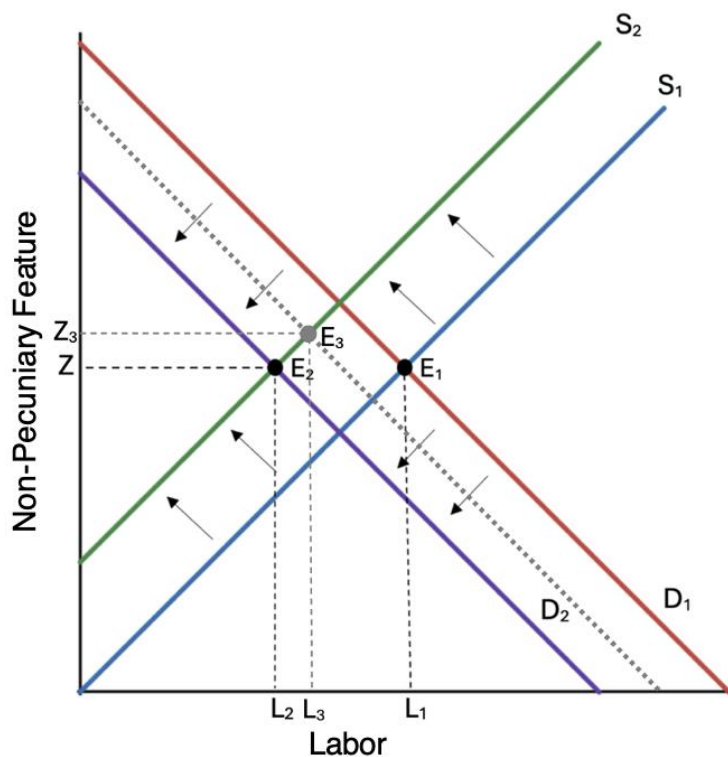
Implications of Dobbs

The overturning of Roe introduces a geographic preference shock. Programs in states with abortion bans might see a decline in their competitiveness due to the reduced utility

they offer. This could lead to either a drop in the number of applicants to these programs or a sorting effect where less competitive applicants fill these positions. Conversely, programs in states without abortion bans or those with protections may become more attractive, increasing their competitiveness. The degree of this shift will likely be more pronounced in programs that were already less desirable based on other features.

The hedonic wage model equilibrium is canonically comprised of a set of touching maxima. To illustrate the basic market dynamics in the context of this paper, consider a simplified depiction of the labor market for providers in states with abortion restrictions depicted in Figure 1, where the initial amount of any desirable non-pecuniary job feature Z , and labor supply L_1 , clear the market at equilibrium point E_1 . S_1 then represents the supply of residents to programs in restrictive states before the restriction is adopted, where the shock of the restriction affects resident geographic preferences such that that the post-policy labor supply is S_2 . That is, simultaneously, programs anticipate the shock and reduce their quotas for new residents, resulting in demand moving from D_1 before policy adoption to D_2 after, yielding a new equilibrium E_2 comprised of a reduced labor supply for the same amount of Z offered.

Theoretical Response of the Market for New Physicians Following the Adoption of an Abortion Restriction



A key non-pecuniary job feature is a program's willingness to take on a wider breadth of applicants. Applicants who received their degree from a US institution are typically more competitive in the match, so programs could widen their preferences for non-US trained physicians (IMGs), thereby shifting the demand curve to the right, resulting in a new equilibrium, E_3 , characterized by labor supply L_3 and non-pecuniary job feature Z_3 , recuperating some of the losses to labor supply, but at a higher cost.

Conclusion

In summary, this model predicts that policy changes affecting geographic preferences will significantly impact the residency match process. Programs in states with abortion restrictions may need to enhance other features or lower their hiring standards to compensate for the loss in geographic desirability, while those in non-ban states may

see increased applicant interest and competitiveness. This adapted hedonic model provides a framework to understand how external shocks can influence applicant behavior in a match-based labor market and makes a novel contribution by extending the model to markets where employers do not compete on wages.

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